

Predicting CSE Undergraduate Students' Careers Using Machine Learning Algorithms in the Context of Bangladesh

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Abstract—Selecting a suitable career path is often difficult for Computer Science and Engineering (CSE) undergraduates because of the wide range of available fields, including software development, data analysis, cybersecurity, and government jobs. This paper proposes a machine learning-based approach to predict suitable career paths with probability by analyzing students' academic performance, technical skills, soft skills, and career preferences. It uses records from 468 CSE undergraduate students with 24 features collected through a structured questionnaire survey, including CGPA, soft skills, academic skills, research ability, publication record, project experience, and multiple preferred career options. Ten classification algorithms were evaluated and compared using accuracy and execution time. The XGBoost Classifier provided the best accuracy of 61.70%, thus earned the status of the most efficient model in predicting careers in this dataset. The analysis also shows that software development is the preferred sector among students. This research analyzes student data to provide relevant career information and improve employability, which will help students select the appropriate career for their bright future. Furthermore, the study assists students to find suitable jobs that align with their interests, abilities, and skill related pursuits

Index Terms—Machine Learning, Career Prediction, Multi-Label Classification, XGBoost, Skills Assessment, Data-Driven Counseling.

I. INTRODUCTION

Every year a large number of students complete their graduation from Computer Science and Engineering (CSE) in Bangladesh. According to a recent study conducted by the Asian Development Bank (ADB), around 20,000 fresh Computer Science and Engineering (CSE) students graduating every year are failing to meet the annual demand for 7,500 - 8,000 new technical personnel. More than 80% of job seekers fail written tests on basic coding skills, mathematics,

and English. There are some reasons behind failure where career guidance is most important [9]. After graduation, most students become confused when it comes to determining one's self in terms of area of interest and self-abilities to choose the right careers. Since today there are multiple career options and competition in the job market.

Many CSE students find it difficult to select a career that fits their interests and skills, which is why it is important to have a data-driven guidance system. This paper proposes a machine learning-based model that predicts the right career paths in line with the performance of students in their studies, technical and soft skills alongside their favorite job variables. Previous studies have attempted to connect individual competencies to career outcomes with several machine-learning methods, and the present research is based on the past studies but provides a more formalized and detailed framework.

Recent studies have expanded career prediction research by using NLP models, hybrid ensembles, and Digital Twin systems. For example, an NLP-based system used SVM, CNN, and LSTM to extract meaningful patterns from student-written text [8]. Another study applied a Digital Twin approach that combined academic results, skill ratings, and sensor-based activity data to support behavioural modeling [5]. Skill-focused models using XGBoost, SVM, and ensemble classifiers have also shown good results when technical skills are included [6], [17]. Exam-based frameworks have also performed well when national assessment scores were used [12]. However, most of these methods still rely on small or artificial datasets and produce only one career prediction for each student. It is obvious that we require an improved multi-label system that makes use of actual student data.

We intend to develop a career prediction machine-learning model among CSE students. In order to create a reliable

system that accurately represents the skill profiles of 468 Bangladeshi students, we worked with their data. Along with influencing factors and preferred career choices, the dataset contains soft skills like problem-solving and teamwork as well as technical skills like cybersecurity, machine learning, and data analysis. In order to determine the best model for multi-label career prediction, the study assesses several machine-learning algorithms. It also seeks to fill gaps of previous work by utilizing a multi-label classification tackling for students pursuing many career paths as well as broader influencing aspects in the widespread effects of job security or career satisfaction, which are commonly important factors for a more holistic and realistic prediction model.

To predict careers for CSE undergraduate students, this research uses a quantitative strategy using Machine Learning models. Using a structured questionnaire data was collected from 468 undergraduates, involving 24 features including CGPA, technical skills, soft skills, publications, projects, job preferences and other aspects that encompass areas such as job security. After preprocessing the data (cleansing, coding, normalizing) a variety of 10 different machine learning algorithms were tested with this dataset. We divided the dataset to form training and testing sets and the models were evaluated based on accuracy where XGBoost had the highest accuracy (61.7%). It offers personalized career suggestions because it matches students' skills and preferences to career opportunities.

Previous research have focused primarily on single-label prediction and used very limited sample sizes (i.e., typically only one possible career path), which do not adequately represent the diversity of potential career paths available to CSE undergraduate students in Bangladesh. We present a multi-label system based on actual student data obtained across Bangladesh in order to close this gap. In contrast to previous studies that only predicted one possible career path, we will suggest several relevant career sectors based on their skills, preferences, and performance. The suggested approach provides a more realistic and comprehensive model of offering data-driven career advice. **To the best of our knowledge, this study is one of the first multi-label career prediction model on Bangladeshi CSE undergraduates based on real institutional data.**

II. KEY CONTRIBUTIONS

The current study provides a number of novel, substantial contributions to data-driven career prediction for CSE undergraduates. The primary contributions are as follows:

- We propose among the first multi-label career prediction frameworks for CSE undergraduates in Bangladesh, addressing the limitation of earlier single-label approaches.
- We put together a real dataset of 468 students from 47 universities, where we collected 24 different academic, technical, behavioural and preference-related features through a structured survey.
- We evaluate 10 machine learning classifiers such as Decision Tree, Random Forest, Logistic Regression, Gradient

Boosting, KNN, XGBoost, SVM, Neural Network and a voting ensemble model.

- We show that XGBoost is a reliable option for multi-label career prediction since it achieves the best overall performance in terms of accuracy, precision, recall, micro/macro F1 scores, and Hamming Loss
- We publicly release the dataset, preprocessing scripts, and full experimental pipeline to support transparency, reproducibility, and future research in data-driven career guidance.

LIST OF ACRONYMS

- CSE - Computer Science and Engineering
- ML - Machine Learning
- XGBoost - Extreme Gradient Boosting
- XGBC - XGBoost Classifier
- DT - Decision Tree
- RF - Random Forest
- GB - Gradient Boosting
- SVM - Support Vector Machine
- SVC - Support Vector Classifier
- KNN - K-Nearest Neighbors
- NN - Neural Network
- HVC - Hard Voting Classifier

III. RELATED WORK

There are numerous approaches to career guidance and more and more recent literature research suggests machine-learning as an effective method of assisting students in making the right career decisions. Previous research was usually capable of proposing only a single option that relied on examination scores or simple tests of interest, neglecting the fact that there are numerous abilities and interests in CSE students. Several recent papers have introduced more features and more advanced models, but some of the key problems, including small datasets and restricted predictions, are still present.

Most previous studies followed single-label approaches. In an early study, the researchers used K-Star, SVM, and Naïve Bayes on interest-based data, and the model ended up giving a little over 89% accuracy [18]. Pandey and Maurya [13] studied the performance of Decision Tree, Random Forest and XGBoost in relation to engineering students. Their soft-voting ensemble achieved 53.03% accuracy using mostly academic grades. Parvez et al. [14] used Naive Bayes, J48, Bagging, and Random Forest to predict IT professionals in Bangladesh, with the last one recording the highest accuracy (67.4). In a recent study in Nigeria [12], the national examination scores, the interests, and extracurricular performance were utilized and an accuracy of 85.7% was achieved with XGBoost. Still, the work was based on a limited feature set and only suggested one career path per student.

Besides these studies, a number of recent studies have investigated various machine-learning methods of career and employability prediction. Kumar et al. [10] used the Logistic Regression, the Random Forest, and the XGBoost with the use of academic scores, internships, certifications, and aptitude

indicators to categorize the students into Placed/ Not Placed, with the highest accuracy of XGBoost being 95.34%. In an Intelligent Career Guidance System, KNN with K-Means was used to assign engineering departments based on a small set of technical and analytical skills, but the manually built dataset and narrow scope reduced its generalizability [3]. Models based on an ensemble e.g. Random Forest, Decision Tree, and Voting Classifiers- have also been investigated, with the as high as 89% accuracy on Kaggle data [16], but constrained by single-output recommendations and unreal world data of an institution. Hybrid deep-learning methods combining RNN, LSTM and NLP are also capable of a very high accuracy (up to 99.33%) when ranking career domains [11], however, such systems are also dependent on a questionnaire-based input of text and do not feature a structured academic representation of skills. Previous research tends to work with small data, subjective skill assessment, and single-label results. This makes it evident that there is a distinct necessity to have a multi-label, skill-based model based on actual student data, which the current work offers.

The other direction of research is on employability prediction, but not career role assignment. Vadivu and Sornalakshmi [21] categorized employability of students with KNN and Naïve Bayes with accuracies of 95.33% and 97.67% respectively. Casuat et al. [2] evaluated six machine-learning models basing on academic records, training scores and interview data, with SVM obtaining 91.22% after using SMOTE. According to Sinha et al. [19], SVM, Decision Trees and Random Forest were applied to predict job readiness depending on the interests, abilities, and academic status. They focus on performance-based features, but they usually leave out soft skills, project experience, research activity, and job preferences all of which are important for meaningful career guidance in CSE.

A number of researches have examined competence-based forecasting systems. Gavhane et al. [6] created the model on the basis of SVM, XGBoost and hybrid ensembles to categorise students into 15 career groups according to technical skills. Reddy [17] suggested a comparable program on the CSE pupils on the basis of XGBoost and Decision Trees, which is based on self-assessed programming, database and algorithmic ability. Although these methods are useful, they still produce only single-label outputs and include very limited behavioural or experiential factors such as teamwork, communication, or research involvement.

Newer systems have NLP and hybrid structures. A study conducted using NLP utilized SVM, CNN and LSTM to predict text written by students with 88.63% accuracy [8]. Digital Twin Based systems consist of academic history, competency scores, sensor data to predict careers and identify activities, training LR, RF, Gradient Boosting and CNN, LSTM, feed forward networks [5]. These systems tend to use small or artificial datasets, regardless of their sophisticated design, and to solve career prediction as a single-label problem.

The available literature demonstrates that even the more sophisticated models have yet to achieve a number of relevant

points. Research on technical skills in the literature [6], [17] that emphasizes career skills in particular fields of study also does not pay attention to behavioral competencies, experience in the field of research, and the area of work in which students want to work. The works centered on employability, such as the work of [2], [21], predominantly use academic scores in their analysis and do not fully capture the scope of the elements determining CSE career choices. Even ensemble and hybrid models, which are still used to predict a single career path, can have overlapping skills that fit several jobs [13], [14]. In many cases, the datasets are small or limited to a single institution, reducing generalizability. These loopholes show the necessity of the realistic multi-label model which is able to take into account various skills and offer more flexible career advice.

To fill these gaps, this paper proposes a multi-label, skill-based career prediction model constructed on actual data gathered on 468 CSE students in 47 universities in Bangladesh. The dataset has 24 academic, technical, behavioural, preference-based, and research-related features, which provide a full picture of the student abilities. The model unlike the previous single-label models, predicts multiple career paths that each student should have based on his skill profile. Ten machine-learning models were compared, and the XGBoost was the most successful in general. All data and workflow are open-source to facilitate reproducibility and further studies.

As observed in comparative studies in Table I, most current career-prediction systems are based on small or self-reported datasets, operate on limited features and nearly always give single-label results. Multiple pieces of work indicate that they can be used with high accuracy with classical machine-learning or deep-learning models, but because of that their scope is restricted, and they are designed to work only with a single output, which cannot be translated into real-world use. To conduct research into these gaps, the research relied on a larger and multi-institution sample of 468 CSE students and proposed a multi-label career recommendation framework that is capable of capturing the overlapping skill sets around CSE undergraduates.

IV. PROPOSED WORK

Existing career guidance platforms generally rely on limited indicators such as academic results or student interests. However, CSE undergraduates often possess diverse and overlapping skills that may correspond to several career fields simultaneously. Due to such a limitation, conventional systems often do not align students with the areas that their real competencies can be more effectively applied.

In order to fill this gap, We propose a multi-label career prediction model incorporating academic performance, technical skills, soft skills, influencing factors, and desired job types. The model shows many possible career paths, rather than one, that match the skills set of a student, which is why this approach is more effective in counseling students.

TABLE I: Condensed comparison of recent employability and career prediction studies.

Study	Main Models	Best Result	Key Limitations
Employability ML (2025)	LR, RF, XGBoost	95.34% (XGB)	Binary only; small dataset
Career Guidance (2022)	KNN, K-Means	>90%	Self-rated skills; no multi-label
Career Prediction (2021)	XGBoost, DT	XGB best	Single output; CS-only
Ensemble Learning (2022)	RF, DT, Voting	89% accuracy	Kaggle dataset; single label
Hybrid DL Model (2023)	RNN, LSTM, NLP	99.33%	Unknown dataset; overfitting risk
Undergrad ML Study (2021)	DT, RF, XGB	53.03% (Voting)	Very small dataset
Student Career ML (2023)	LR, SVM, RF	RF best	No accuracy reported; no multi-label
Nigeria Career (2025)	LR, RF, XGB	85.7% (XGB)	Country-specific; no multi-label
SVM+SMOTE (2020)	NB, KNN, SVM	91.22% (SVM)	Binary only; subjective scoring
Digital Twin (2025)	LR, RF, CNN, LSTM	98% (LR)	Single-label; complex setup
IT Personnel BD (2017)	NB, J48, RF	67.4% (RF)	Small dataset; low accuracy
NLP Career System (2024)	SVM, CNN, LSTM	88.63% (SVM)	Text-only; no multi-label

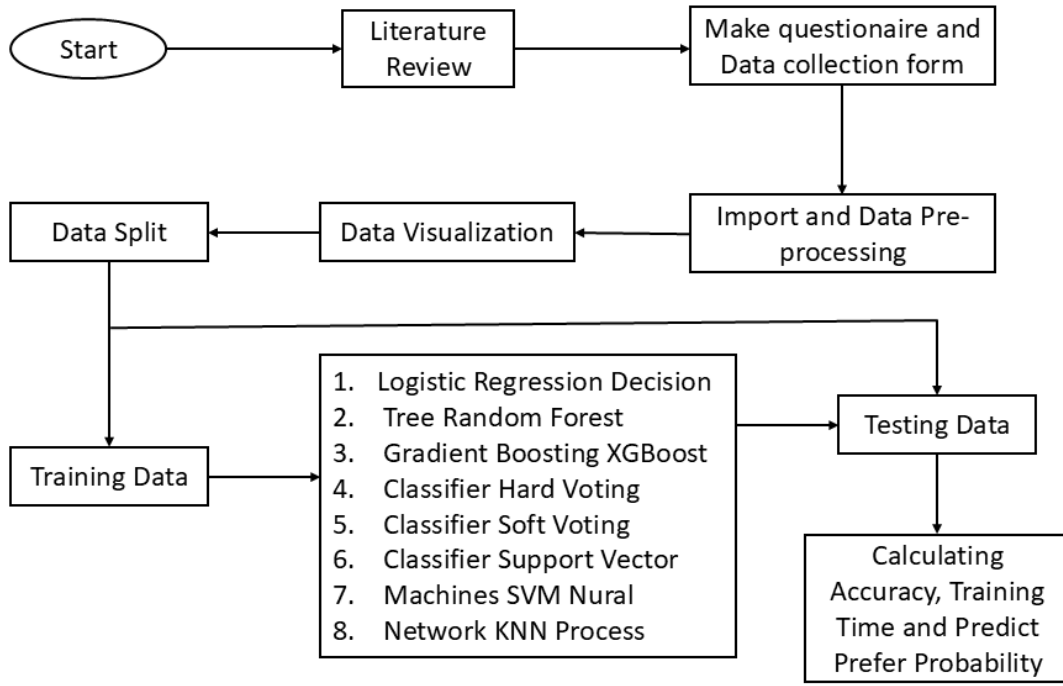


Fig. 1: Workflow of the proposed machine-learning based career prediction system for CSE undergraduates.

V. RESEARCH DESIGN AND METHODOLOGY

A. Data Collection

A Web-based survey platform was used to create a structured questionnaire to gather data on the skills of the students, their academic performance, and the preferred fields of their career. The survey was hosted online ¹ and remained active from 24 April 2024 to 21 November 2024. A total of 468 responses were collected from students enrolled across 47

public and private universities in Bangladesh. All responses were recorded and stored in a Google Sheet for further processing.

The complete dataset used for the analysis, including the raw (anonymized) responses, is publicly available for reproducibility². All associated source codes, preprocessing scripts, and visualization notebooks are accessible in the project's GitHub repository³.

¹Survey-form: <https://israt241thesis.blogspot.com/2024/08/april-01-2024-prediction-of-key-factors.html>

²Dataset: <https://github.com/IsratJahanR/CSE-Thesis/blob/main/CSE%20Thesis%20-%20Sheet1.csv>

³Code repository: <https://github.com/IsratJahanR/CSE-Thesis>

The survey included 24 attributes and five broad categories, i.e., Personal Information, Soft Skill Ratings, Academic Skill Ratings, Influencing Factors, and Preferred Job Categories. Table II shows the question items of Personal Information category.

TABLE II: Survey questions and corresponding response categories for personal information.

Survey Question	Students' Response
Gender	Male / Female / Other
CGPA	Write current CGPA
Year	1st / 2nd / 3rd / 4th
University Name	GSTU

Section 2 contains soft skill ratings, which consist of 4 questionnaires named critical thinking, problem solving, teamwork, and communication skills. Students rated their skill levels on basic, intermediate, or Advanced.

There was a three-point scale- Basic, Intermediate or Advanced that was used to assess academic skills. Students also reported their number of publications and completed projects. The following are the areas of academic skills that were used in the survey:

- 1) Software Engineering Principle
- 2) Data Structure and Algorithm
- 3) Database Management
- 4) Data Analysis
- 5) Web Development Skills
- 6) Understanding of Computer Architecture
- 7) Understanding of Operating Systems
- 8) Networking Concepts
- 9) Cybersecurity
- 10) Machine Learning
- 11) Robotics
- 12) Research Skills
- 13) How many publications?
- 14) How many project?

Influencing factors capture the motivations that shape students' career choices. The characteristics assessed are as follows:

- 1) Social Acceptance: Refers to the importance students place on being valued and recognized within their social or professional circles.
- 2) Job Security: Highlights the desire for stable and long-term employment opportunities.
- 3) Job Environment: Captures the significance of a positive and conducive workplace environment for students.
- 4) Salary: Reflects the role of competitive and rewarding compensation in shaping career choices.
- 5) Flexible Work-Life Balance: Emphasizes the value of achieving harmony between professional responsibilities and personal life.

These responses highlight which motivations students prioritize when choosing a profession.

The Preferred Job part was added to get a better idea of the career interests of the students. The list of potential job opportunities was given in a comprehensive manner giving students the opportunity to choose several options, which was similar to their career interests. The available options are listed below:

- | | |
|----------------------------|--------------------|
| 1. Software Development | 10. Management |
| 2. Networking Sector | 11. Teaching |
| 3. Cyber Security | 12. Researcher |
| 4. Database Administration | 13. Bank Job |
| 5. Data Analysis | 14. BCS |
| 6. UI/UX Designing | 15. Business |
| 7. ML/AI Engineer | 16. Government Job |
| 8. Hardware Sector | 17. Abroad |
| 9. Gaming | 18. Others |

These responses provide insight into the diversity of students' career goals across technical and non-technical domains.

B. Data Preprocessing

The raw survey data had empty records, categorical data and a number of formatting anomalies that needed to be converted before model training. The optional fields contained missing values that were replaced with either zero or an unrated label such that no information was eliminated in preprocessing. To maintain numerical consistency and prevent scale-related bias, continuous attributes were standardized using the *StandardScaler*. The data was cleaned, organized and was fit to be analyzed through these pre-processing actions.

C. Feature Reduction

The initial dataset included a number of non-informative fields such as timestamp, email address, and descriptive text entries. These attributes are not useful in career prediction and might introduce noise thus they were eliminated to retain only the academic, technical, behavioral as well as preference-related attributes required by the model.

D. Encoding Categorical Features

Since the dataset primarily consisted of categorical variables, all features were transformed into numerical representations prior to model training. Skill-related features that follow an inherent progression were encoded using an ordinal scale:

$$x_{\text{skill}} = \begin{cases} 3, & \text{Advanced} \\ 2, & \text{Intermediate} \\ 1, & \text{Basic} \\ 0, & \text{Unrated} \end{cases}$$

Multi-select attributes, including influencing factors and preferred job categories, were encoded using the *MultiLabelBinarizer* to generate multi-hot vectors that represent all selections made by each student. This ensured that output labels and input attributes were structured appropriately for multi-label machine-learning models. This encoding ensures ordinal relationships are preserved during training.

E. Data Visualizations

Data visualization is the process of finding trends and correlations in our data by representing it pictorially. Using data visualization, we can get a visual summary of our data and also help to describe stories to the users by organizing the data into a form easier to understand.

In Figure 2, the results indicate that the software careers have most preferences since 228 students (48.73%) opted to Software Development much higher than other alternatives. The next opportunities include abroad (81 students 17.3%), ML/AI Engineering (15.6%), and Data Analysis (15.2%). Government Jobs (13.7%) and BCS (9.6%) are more acceptable, although creative options like UI/UX Designing (3.2%) and Gaming (2.1%) are also somewhat of a niche. On the whole, the statistics indicates that there is a high disposition towards technical and programming-oriented positions.

Figure 3 shows the distribution of CGPA and core soft skills, where most students rated themselves as Intermediate or Advanced, indicating a generally strong foundation. Figure 4 illustrates representative technical skill distributions, revealing that while students perform well in fundamental subjects, their proficiency drops in advanced areas such as Networking, Cyber Security, and Machine Learning. This highlights a clear gap between basic understanding and higher-level competencies. Only selected skills are visualized due to space limits, and the remaining skills follow similar patterns.

As shown in Figure 5, students complete significantly more projects than publications, with projects showing a wider spread around the median while publications remain minimal. Figure 6 indicates that technical careers such as software, ML/AI, database, cybersecurity, and networking are mainly driven by salary, whereas job security and social acceptance influence preferences for government, BCS, banking, teaching, research, and abroad careers. Less popular fields such as UI/UX, gaming, and hardware are influenced by both salary and work environment factors.

The dataset was large, which made features reduction methods relevant to eliminate variables that made no significant contribution to prediction. This assists in enhancing the accuracy of the model and training is more effective. To reveal the relationship of the rest of the features with each other, a correlation heatmap was created, with the closer the value is to +1, the stronger a positive relationship and the closer the value is to -1, the stronger the negative relationship.

Figure 7 shows the interrelation between the different columns in the dataset. It established a strong positive relationship between the research skills and the publication which shows that research skills are positively related with publication. Next only to this, the results for data analysis and machine learning skills show that these two pairs are related with each other. Moreover, web development and software engineering are ranked third in terms of the connection between them with considering that these spheres are closely related.

F. Model Preparation

The fact that various low-frequency career categories had to be involved resulted in the need to merge the related job areas in order to eliminate the imbalance in the classes and enhance model generalization. In order to maintain meaningful structure of the merged categories, the process of grouping was informed by a set of skill similarity and student preference co-occurrence as well as association rule mining.

BCS and Bank Job were both combined in a single category, which was called the Government Sector. The association rule mining (Table III) shows that, there was always high co-occurrence of these three preferences whereby the confidence values were above 0.70 and lift values were above 5.0. indicating strong co-occurrence patterns [1]. It means that students who chose one of these roles had a high probability of choosing the rest of them. Furthermore, the three jobs have close entry requirements, high focus on job security, and close preparation channels, and thus, their merging will be statistically and semantically correct.

TABLE III: Association rules among Bank Job, BCS, and Government Job preferences.

Antecedents	Consequents	Support	Confidence	Lift
Bank Job	BCS	32	0.71	7.39
BCS	Bank Job	32	0.71	7.39
BCS	Govt Job	36	0.80	5.85
Bank Job	Govt Job	36	0.80	5.85
Bank Job, Govt Job	BCS	29	0.80	8.37
Bank Job, BCS	Govt Job	29	0.90	6.62
BCS, Govt Job	Bank Job	29	0.80	8.37

Likewise, job positions involving software-like Software Development, Data Analysis, ML/AI Engineering, Database Administration, Cyber Security, Gaming and UI/UX Designing were combined into the *Software Sector*. The disciplines demand technical skills that overlap like programming, algorithmic thinking, system-level knowledge and analytical skills. The multi-selection across these categories by the students also exhibited high levels of multi-selection which validates the perception by CSE undergraduates that these roles belong to a common skill ecosystem. The combination of them thus capturing their inherent interdependencies makes the multi-label learning process stable.

All the attributes were scaled to make sure that no one attribute had unreasonable impact on the learning process. Then the models were trained using the processed data divided into a training and testing portion in an 80:20 ratio to enable fair assessment.

G. Classifiers

In this research, we have selected 10 popular machine learning and ensemble learning classification algorithms. These are

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Voting Classifier
5. Gradient Boosting
6. XGBoost Classifier
7. Hard Voting Classifier
8. Soft Voting Classifier
9. Multilayer Perceptron
10. K-Nearest Neighbors

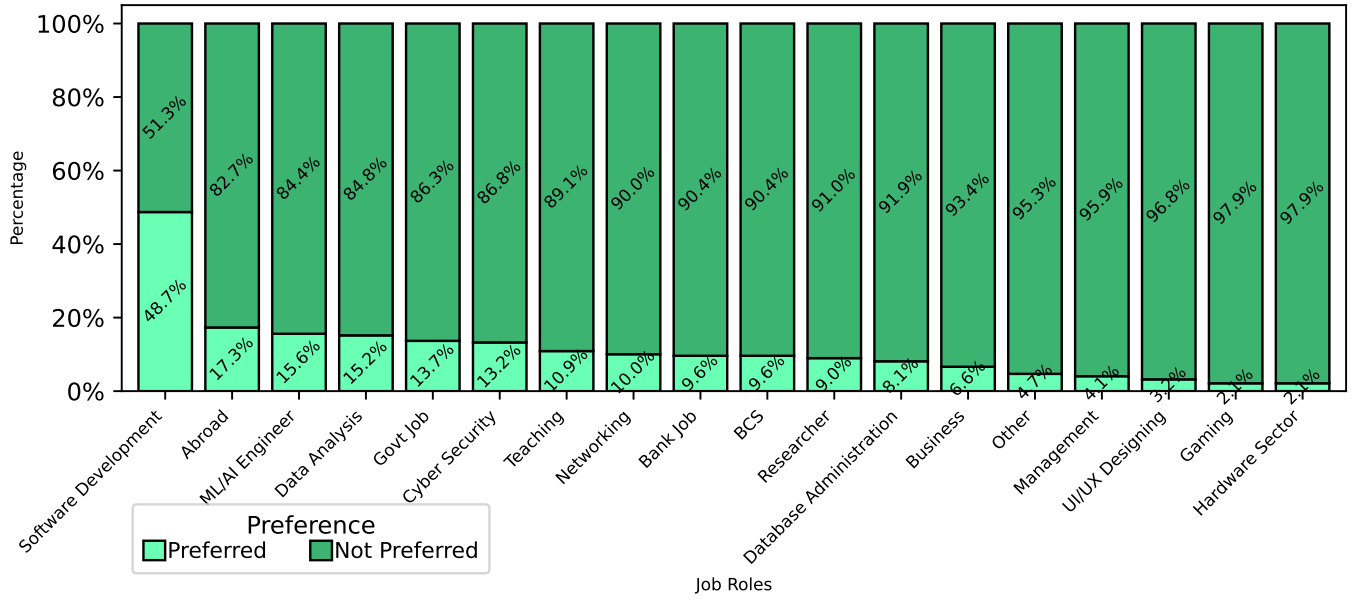


Fig. 2: Distribution of preferred job roles among 468 CSE students.

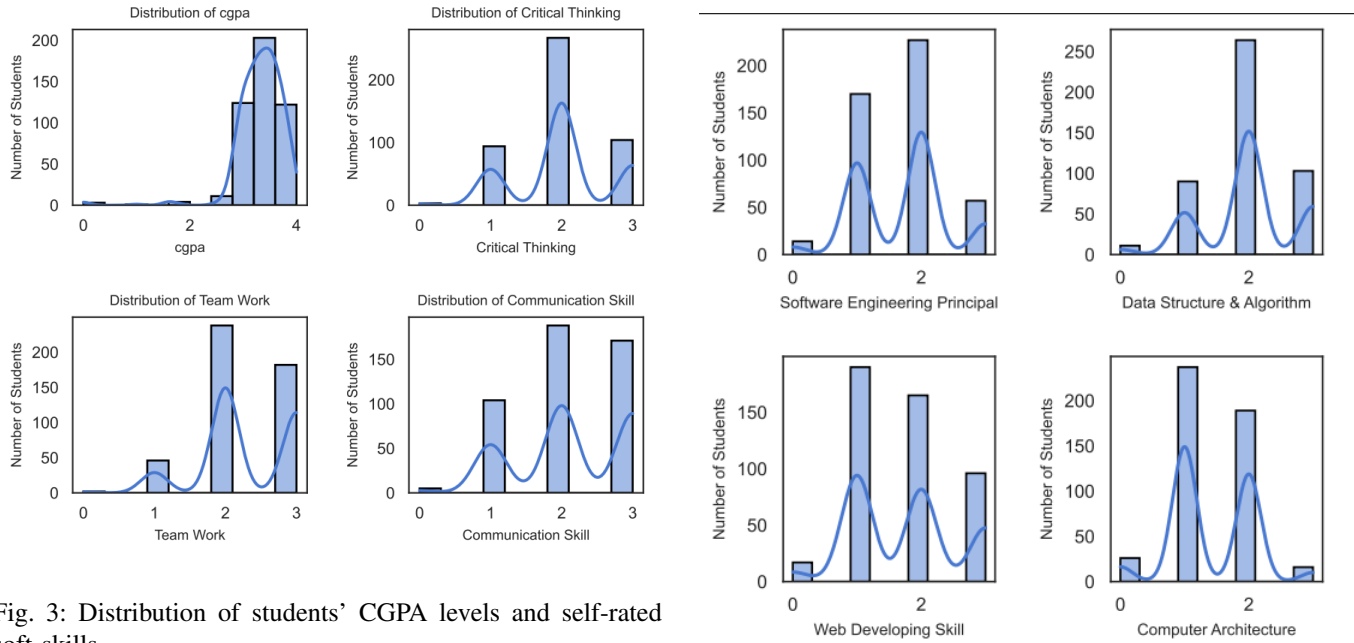


Fig. 3: Distribution of students' CGPA levels and self-rated soft skills.

VI. RESULTS

A. Accuracy

Now that we have prepared our algorithm, the achieved accuracies are provided in the following Table IV where accuracy of each algorithm is defined how much time it consumed to make itself ready i.e., training time. Each algorithm was tuned using some hyperparameters to obtain stable and similar performance across models.

Though the overall accuracy values seem to be lower as

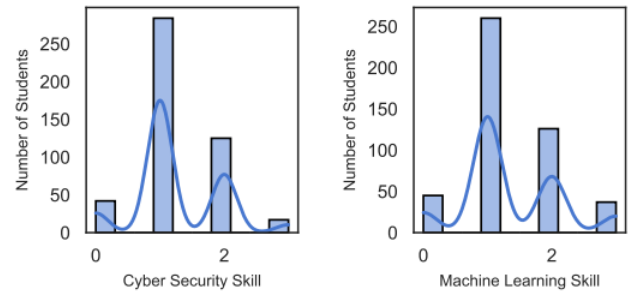


Fig. 4: Technical Skills Distribution.

TABLE IV: Performance comparison of ten machine-learning classifiers based on accuracy and execution time.

SI	Algorithm	Hyperparameter	Accuracy (%)	Execution Time (s)
1	LR	max_iter = 1000	42.55	0.09
2	DT	max_depth = 10	37.23	0.08
3	RF	random_state = 42	47.87	1.57
4	GB	random_state = 42	51.06	1.71
5	XGBoost	learning_rate = 0.1, n_estimators = 200	61.70	0.75
6	HVC	voting = hard, max_iter = 10000	51.06	1.80
7	SVC	voting = soft, max_iter = 10000	51.06	1.79
8	SVM	kernel = linear, random_state = 42	43.61	0.37
9	NN	hidden_layer_sizes = (100, 50), max_iter = 1000	47.87	3.47
10	KNN	n_neighbors = 7, weights = distance	55.31	0.09

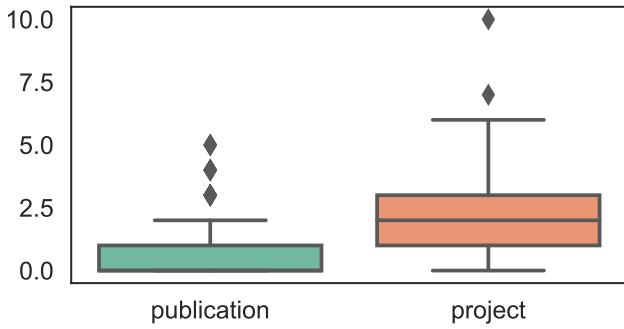


Fig. 5: Boxplots comparing student project counts and publication counts.

compared to those of the standard single-label classification experiments, this is to be expected in multi-label prediction experiments. In our dataset, any student may be a member of multiple categories of careers at a time, which complicates the boundary of the decision and decreases the discriminative ability of accuracy as an evaluation measure. Moreover, the career labels are rather balanced, and numerous students are similar in the mid-range skill boxes, which leads to duplicated classes. Based on these facts, accuracy can not give a complete picture of model performance and multi-label metrics which include precision, recall, micro/macro F1 scores, Hamming Loss is a better assessment in in multi-label settings [20].

The results of the ten classifiers are summarized in Table IV based on their performance and time. XGBoost offers the best overall performance since it has the best accuracy (61.70 percent) at a moderate training time (0.75 seconds). KNN offers the best second-ranking (55.31 percent) and the worst-ranking execution time (0.09 seconds) which means it is highly executed computationally. The hard/soft voting ensembles and Gradient Boosting have wider accuracies of about 51 percent but take a longer time to train. With the complexity of the multi-label task, Logistic Regression and Decision Tree demonstrate significantly lower accuracy, but they are trained very fast. The Neural Network is medium accurate with the

highest cost of execution (3.47 seconds). In combination with Figure 8, these findings demonstrate the trade-off between predictive performance and computation, and XGBoost provides the most balanced and effective solution to multi-label career prediction.

B. Additional Evaluation Metrics

Because it is not possible to evaluate only by accuracy, other measures of evaluation were taken into consideration and it includes Precision, Recall, Micro F1, Macro F1 and Hamming Loss. Table in Appendix B summarizes these findings in Table V. On the whole, the models have different strengths, yet XGBoost has the most balanced and consistent results in all metrics, especially in Recall, Micro F1, and Macro F1, and also the lowest Hamming Loss.

TABLE V: Advanced performance metrics for all classifiers.

Model	Precision	Recall	F1(Micro)	F1(Macro)	HL
LR	0.811	0.625	0.703	0.551	0.081
DT	0.671	0.689	0.680	0.525	0.100
RF	0.910	0.620	0.737	0.517	0.061
GB	0.888	0.648	0.749	0.589	0.067
XGBoost	0.856	0.737	0.792	0.695	0.060
HVC	0.852	0.650	0.767	0.651	0.062
SVC	0.861	0.665	0.761	0.651	0.063
SVM	0.716	0.577	0.677	0.531	0.082
NN	0.799	0.627	0.697	0.579	0.080
KNN	0.876	0.634	0.736	0.538	0.070

In each and every model, the Precision is maximized in case of the Random Forest and Gradient Boosting which means that there are lesser numbers of incorrect predictions. XGBoost however is the best in Recall (0.737) and indicates that it can capture more true labels, hence significant in a multi-label environment. Figure 9, F1 Micro-Comparison further demonstrates that XG Boost is the most powerful in terms of overall performance and Hard Voting, Soft Voting, Gradient Boosting, and KNN are directly below. Other models like Logistic Regression, SVM, and Decision Tree have worse performance because of the inability to deal with overlapping skill distributions.

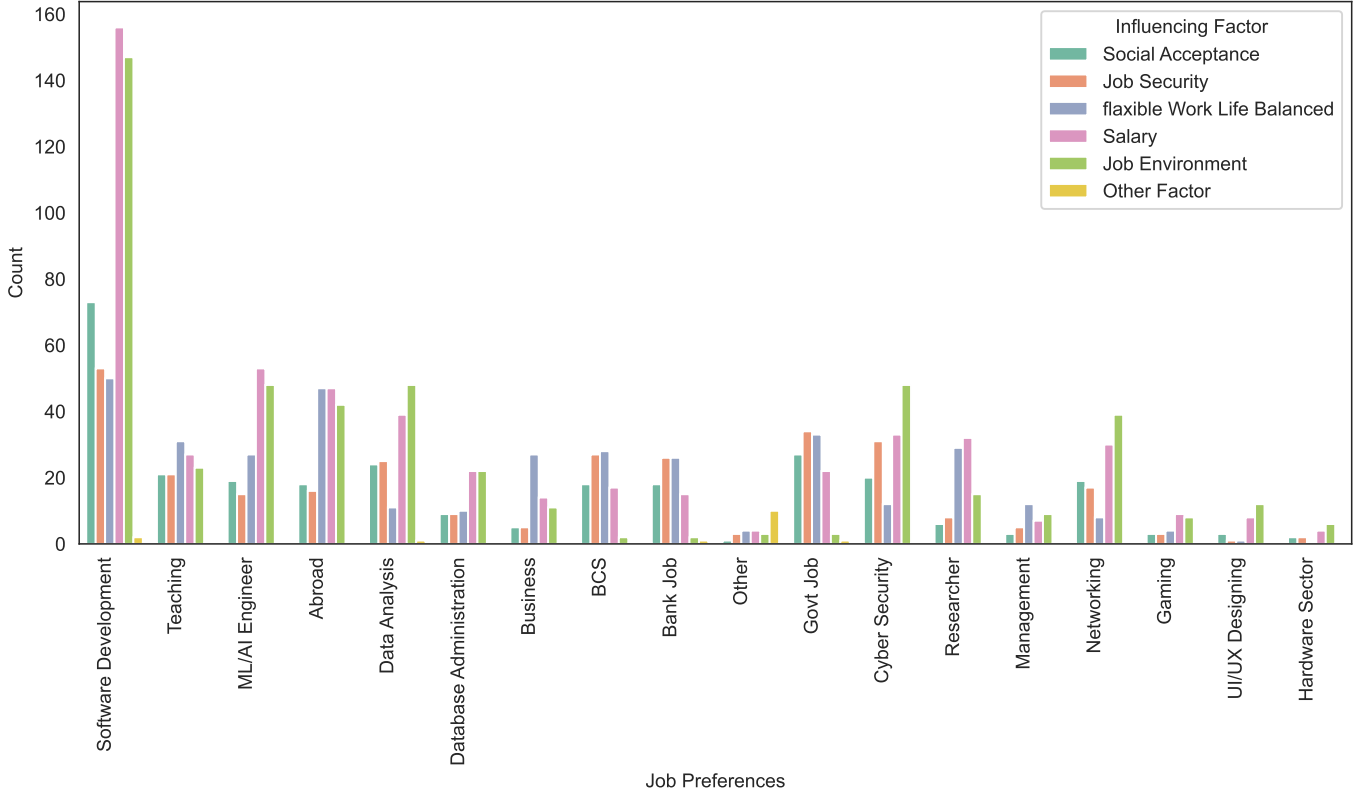


Fig. 6: Influencing factors associated with students' preferred career choices.

This was the results of these findings showing that XGBoost gives the best estimates in both frequent and rare career groups so it is the most efficient model to use in this multi-label classification task.

C. Overall Comparative Interpretation

In all metrics of evaluation measurements, the XGBoost outshines the other classifiers and is more robust, balanced performance, and good generalization ability of multi-label career prediction. Its high accuracy, high recall, good F1 scores, and smallest Hamming Loss are all that make XGBoost the most. effective data-driven career recommendation system model.

D. Test Sample Evaluation

In order to measure the effectiveness of the proposed model in mapping the profile of skills of students to an appropriate career route, two representative samples of CSE undergraduate students were taken. These test samples are shown only to demonstrate the model's prediction behavior and were not used for evaluation. Each sample contains the complete set of raw and encoded skills used as model input, as presented in Tables VI and VIII. The corresponding predicted career probabilities generated by all machine-learning classifiers are shown in Tables VII and IX. This subsection analyzes both the input skill distributions and the resulting model outputs to understand how the system interprets student capabilities.

TABLE VI: Raw and encoded skill values for Test Sample 1.

Skill	Rating	Encoded
gender	male	1
year	3rd	3
cgpa	3.90	3.90
Critical Thinking	Intermediate	2
Problem Solving	Basic	1
Communication Skill	Intermediate	2
Software Eng. Principle	Basic	1
Data Structure & Algorithm	Intermediate	2
Database Management	Intermediate	2
Data Analysis Skill	Intermediate	2
Web Development Skill	N/A	0
Computer Architecture	N/A	0
Operating System	Basic	1
Networking Concept	Basic	1
Cyber Security Skill	Basic	1
Machine Learning Skill	Basic	1
Research Skill	Advanced	3
Publication	5	5
Project	3	3
Influencing Factor	Job Env.	Job Env.

1) *Test Sample 1:* Test Sample VI represents a student with a strong academic background, as shown by a CGPA of 3.90, advanced research skills (Research Skill = 3), five publications, and participation in multiple projects. The student has moderate proficiency in core computing areas such as data structures,

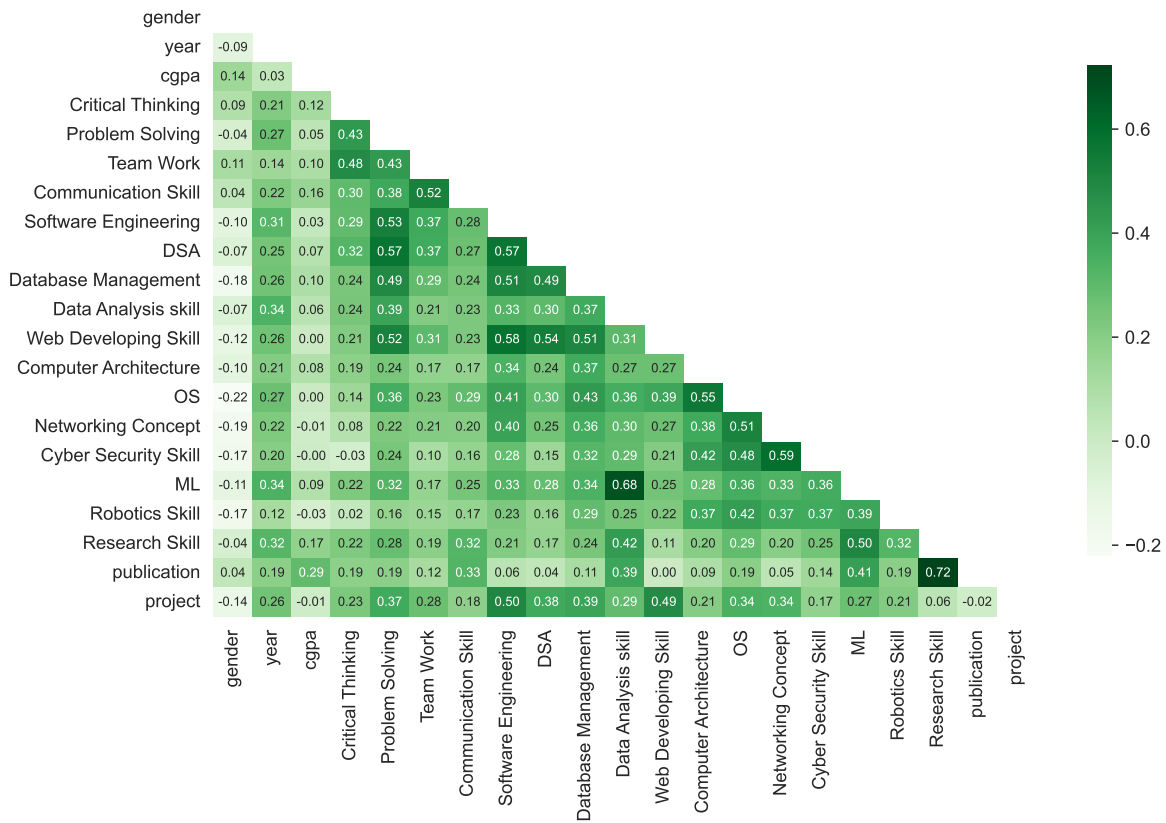


Fig. 7: Correlation matrix showing relationships among skills, academic results, and influencing factors.

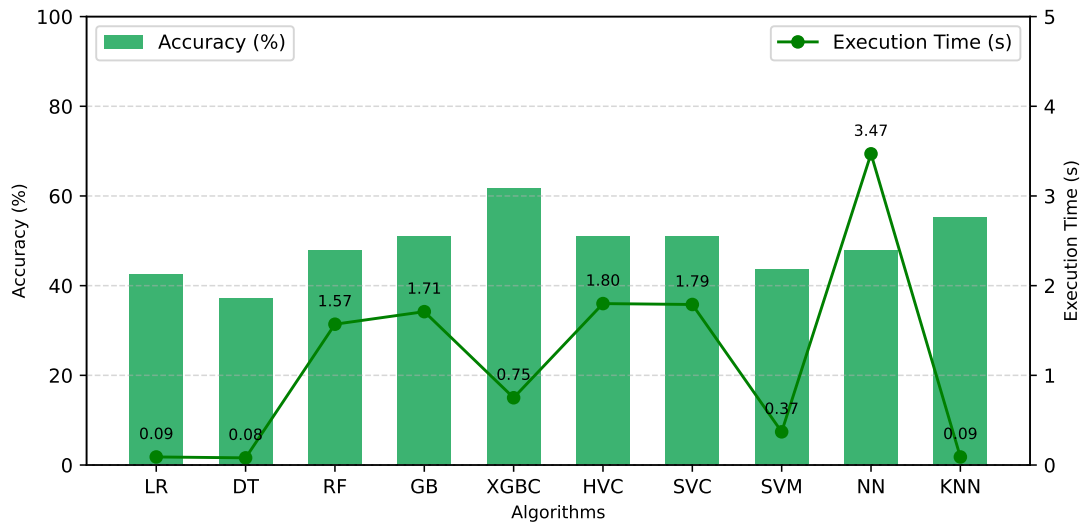


Fig. 8: Performance Measurement using Accuracy and Execution time.

algorithms, database management, and communication skills.

Nonetheless, proficiencies in some applied and specialized domains such as web development, computer architecture, robotics, cybersecurity, and machine learning are either fundamental or not mentioned. This implies that the strengths of the student have more academic oriented and research oriented

strengths than practice oriented strengths.

All models suggest that the most appropriate career option is the one that is called Abroad as it is shown in Table VII. The second choice that seems to be the most relevant option is Researcher that is not surprising given that the student has an excellent research profile. The match of the Software

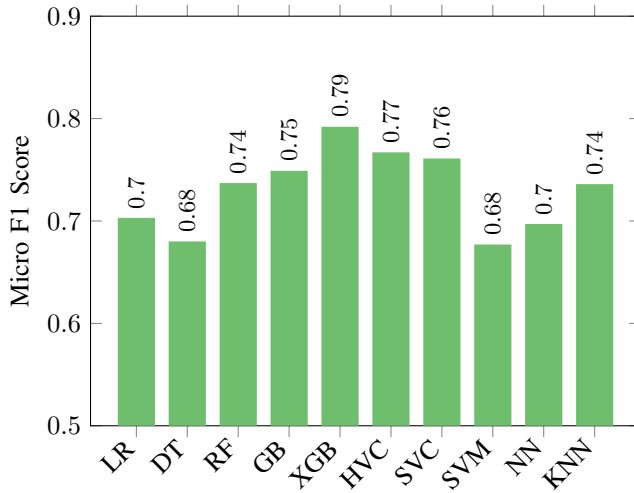


Fig. 9: Micro F1 score comparison among classifiers.

TABLE VII: Predicted career probabilities for Sample 1 across ML models.

(a) Traditional ML Models					
Job	LR	DT	RF	GB	XGBoost
Abroad	99.56	100	56	73.65	90.31
Researcher	93.80	100	22	59.53	31.19
Software Sector	54.18	100	50	14.58	22.73
Networking	29.65	0	10	1.13	0.11

(b) Advanced ML Models				
Job	SVC	SVM	NN	KNN
Abroad	83.90	55.24	99.98	86.23
Researcher	61.30	7.26	0.19	0.00
Software Sector	48.30	69.25	84.46	71.90
Networking	8.18	6.95	0.01	0.00

Sector is moderate, whereas other alternatives like networking, government jobs, hardware, and management are low because they do not have related skills.

2) *Test Sample 2*: Test Sample VIII shows a different skill profile compared to Sample VI. Although the student's CGPA is moderate (3.19), the student demonstrates strong applied technical skills. Higher ratings in problem solving, software engineering principles, web development, and computer architecture indicate a development-focused learning path. In contrast, research ability is low (Research Skill = 1), and salary is identified as the main influencing factor, suggesting a preference for industry-oriented roles.

As shown in Table IX, all models consistently identify the *Software Sector* as the most suitable career option, with predicted probabilities exceeding 96%. This outcome reflects the student's strong software and programming skills. *Abroad* appears as a secondary option, while business roles receive moderate support. Other career paths, including teaching, research, government, and hardware roles, show low suitability due to limited related skills.

3) *Comparative Insight*: The two samples show that the model can clearly distinguish different student strengths.

TABLE VIII: Raw and encoded skill values for Test Sample 2.

Skill	Rating	Encoded
gender	male	1
year	4th	4
cgpa	3.19	3.19
Critical Thinking	Intermediate	2
Problem Solving	Advanced	3
Communication Skill	Intermediate	2
Software Eng. Principle	Advanced	3
Data Structure & Algorithm	Intermediate	2
Database Management	Basic	1
Data Analysis Skill	Basic	1
Web Development Skill	Advanced	3
Computer Architecture	Intermediate	2
Operating System	Intermediate	2
Networking Concept	Basic	1
Cyber Security Skill	Basic	1
Machine Learning Skill	Basic	1
Research Skill	Basic	1
Publication	1	1
Project	4	4
Influencing Factor	Salary	Salary

TABLE IX: Predicted career probabilities for Sample 2 across ML models.

(a) Traditional ML Models					
Job	LR	DT	RF	GB	XGBoost
Software Sector	96.31	100	90	99.02	99.55
Business	11.47	0	16	7.25	0.51
Abroad	9.21	0	31	22.16	11.45
Researcher	4.07	0	0	0.09	0.02

(b) Advanced ML Models				
Job	SVC	SVM	NN	KNN
Software Sector	96.98	92.82	97.78	100.00
Business	7.05	24.44	21.92	26.14
Abroad	14.76	26.79	85.26	30.56
Researcher	0.84	5.60	0.12	0.00

Sample 1 is directed toward abroad and research roles due to strong analytical and research skills, while Sample 2 is matched mainly to software careers because of strong practical programming abilities. This confirms that the model reliably aligns career suggestions with each student's skill profile.

E. Error Analysis

Although the performance of the classifiers is reasonable, several important error patterns were observed during evaluation.

1) **Misclassification of minority classes**: Career categories such as Hardware Sector, Management, and Teaching are represented by only a small number of instances in the dataset. This imbalance makes even strong models mislabel the smaller groups as larger ones like the Software Sector or Abroad. This misclassification is typical in multi-class learning cases where there are imbalanced distributions of labels.

2) **Similar Skill Patterns**: Many career fields share similar basic requirements. For example, Software, Networking, and Cyber Security all require programming and system knowl-

edge. So students who have average skills in several technical subjects often fall into similar areas, and this makes the models produce different results for the same student.

3) Missing Skill Ratings: A significant number of students did not provide ratings for areas such as Web Development, Robotics, Computer Architecture, and Machine Learning. These missing entries caused tree-based models and KNN to shift their predictions toward more frequently occurring categories, reflecting a bias introduced by incomplete data. This shows that the model is quite sensitive to missing skill data, and it would work better if the gaps were handled in a stronger or smarter way.

4) Influence of dominant academic indicators: Students who had a high CGPA, research ability or numerous publications were always put in the category of the Abroad and Researcher. On the other hand, the *Software Sector* was mostly associated with individuals who showed better applied programming skills but lower academic indications. This trend shows that academic qualities have an unduly high impact on various algorithms, especially those with linear decision curves.

These errors show where improvements are needed. Future research ought to be directed towards improved feature design, more efficient treatment of missing data, and an improved balance between career types.

F. Summary of Findings

All the models and test samples point to some common patterns.

1) Model Performance: Among all classifiers, XGBoost demonstrated the highest predictive accuracy (61.70%) and offered the most balanced trade-off between performance and computational efficiency. KNN was the most computationally efficient, accuracy and execution time at the same time. More simply as Logistic Regression and Decision Tree achieved a significant worse result indicating the handicaps in modelling multidimensional skill-career relationships.

2) Test Sample Interpretation: For Test Sample 1, which exhibited high academic achievement, strong research ability, and multiple publications, all the algorithms suggested *Abroad* as the best career choice, then *Researcher*. Test Sample 2, conversely, reflected high applied technical skills and all classifiers agreed that the best career choice was that of the *Software Sector* with a probability of more than 96

3) Influence of Skill Patterns: Analytical and research-oriented profiles identified themselves with career paths whose rationale was academically oriented whereas development-oriented skill profiles identified themselves with software engineering positions. It means that the suggested model has a good capability of differentiating theoretical and practical orientations of skills.

4) Dataset Characteristics: The unbalanced data and missing skill values led to many of the prediction mistakes. However, the general agreement from most of the models is an indicator of strength in the determination of mainstream career pathways.

The framework works well in connecting students' skills with the careers that fit them, and it can be quite useful for counseling and helping students choose their paths.

VII. DISCUSSION

Although the data has limitations, our research proves that the idea of machine learning use in career advice has a high potential. The XGBoost Classifier proved to be the best model because it handles complex student data very well. The analysis suggests that three things are most important for predicting a career: CGPA, Project Count, and Research Skills. This success, even with a small dataset, validates our approach and highlights which student achievements truly matter in this field.

The findings indicate that there is a distinct trend in the way that the models reacted to the data. Models like the Logistic Regression, Decision Tree and SVM, in contrast, were unable to cluster the mid-level skill distributions of many students, which were very similar. This is also supported by the multi-label metrics, which place XGBoost at the top of the Micro-F1 score (0.792) and the lowest Hamming Loss (0.060), which will have a superior stability and reliability across all the career categories.

Another factor affecting performance is the high concentration of students with mid-level proficiency across most technical skills. This causes very similar representations of features, and it decreases the classifiers capacity to differentiate between the tightly clustered categories of career and more overlap in the inferences.

These issues highlight several opportunities for improvement. Future directions would be to improve future work by incorporating methods like SMOTE or class-weighted learning and more robust feature engineering to improve discrimination of the model. Moreover, it is anticipated that by increasing the set of institutions and time, generalizability will be enhanced by the data.

In an applied sense, the findings present some critical trends on how students develop their competencies. The basic skills are moderately established, though the technical skills at an advanced level seem to be consistent and weak, indicating that specific academic interventions are necessary. The universities can use this information to introduce more structured skill-training in more disciplines such as the Machine Learning, Networking, Cyber Security, or research methodology. Even though XGBoost shows promise as a decision-support system, its performance suggests that predictive models are to serve as an adjunct-to, rather than a replacement-of human academic advisors and career counseling models.

In summary, the research indicates the possible and negative aspects of data-driven career guidance systems. Machine learning might be a useful support tool in finding out the right career path, however, to gain higher predictive accuracy more balanced feature distributions, and more coverage of student behaviour, interests, and competencies will be needed.

VIII. THREATS TO VALIDITY

Although the results are encouraging, several threats to validity should be acknowledged to evaluate the robustness of the findings.

The **internal validity** can be influenced by the use of self-reported rating of skills and this gives the chance of the occurrence of recall bias, social desirability bias or different rating scales that were used [15]. Such subjective variations have the potential of affecting the capacity of the model to learn the actual underlying skill patterns. Also, hyperparameter tuning was conducted under restricted computational constraints and this can only allow some models to be tuned to their best configuration.

External validity is limited because the dataset only represents students from one academic field and one region. Their skill levels, coursework, and career interests may not match students from other universities or countries, so the model may not work the same everywhere. Using this framework in a different setting would likely require retraining or adapting the model to that new environment.

Construct validity is concerned with the ability of the selected features to reflect the concept of technical skills, soft skills and career readiness. Although the dataset has 24 attributes, it still does not consider such significant variables as project complexity, internship background, coding sample quality, and objective skill assessments [4]. The model would not be able to use the true capacity of the student without these elements.

Conclusion validity may also be influenced by the moderate dataset size (468 samples) and the imbalance among the ten career categories [7]. These problems may undermine all the statistical power of the findings and bring greater diversity in the performance of the models. Even though we have reported accuracy and multi-label measures, results would be more robust with the application of such techniques as cross-validation or confidence measures.

Overall, these issues show that future studies should use bigger and more diverse datasets, include stronger and more meaningful features, and apply more careful validation to get more reliable results.

IX. LIMITATIONS

There are a number of limitations that should be considered when interpreting the results. First, the dataset size (468 samples) is relatively small for a ten-class multi-label problem, and some career categories contain very few observations. This asymmetry complicates machine-learning models with the ability to start delineating various classes.

Second, the data is based on self-reported skill ratings that can be biased and might not be the perfect indicator of technical capability. Certain middle level ratings and absent values also decrease the feature clarity. Moreover, the data was collected on a voluntary and narrow basis in the universities and thus the data is unlikely to capture the entire population of the CSE students in Bangladesh.

Third, the research based on data at one point in time. Because the skills and career interests of the students can change over time, the long-term data would provide a better understanding of the career development.

Fourth, while multiple evaluation metrics were used, the study does not include real employment outcome data. Career labels reflect students' preferred choices rather than actual job placements, preventing validation against long-term career trajectories.

Despite these weaknesses, the results clearly demonstrate the strong potential of multi-label machine learning models in providing evidence-based academic counseling and establish definite connections for building better future career-prediction systems.

X. CONCLUSIONS

In this study, we implemented a set of machine learning techniques to predict which career paths are most suitable for CSE undergraduate students based on their academic background, technical abilities, soft skills, and personal influencing factors.

The proposed model incorporates a diverse set of input features, including CGPA, communication and problem-solving abilities, knowledge of core CSE courses, project and publication records, and selected career-influencing factors. The output variable consists of ten possible career categories: Abroad, Researcher, Software Sector, Networking, Teaching, Government Job, Business, Management, Hardware Sector, and Others.

We examined the performance of ten widely used machine learning and ensemble classifiers: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, Hard Voting, Soft Voting, Support Vector Machine, Multilayer Perceptron, and K-Nearest Neighbors. Based on predictive accuracy, the top-performing models are **XGBoost** (61.7%) and **KNN** (55.31%). The remaining models showed moderate to lower performance, reflecting the increasing difficulty of modeling overlapping skill-career relationships in a multi-label setting.

XGBoost has the highest predictive accuracy and Decision Tree has the lowest cost of computation of all the algorithms. Neural Network requires the highest training time due to its complex architecture. Based on both accuracy and efficiency, XGBoost would be the most appropriate in this case study to predict career preferences among CSE students. Other systems could only identify a single career whereas in our model, multiple ones can be identified. This reflects the real situation where CSE students often fit into more than one career area.

The results indicate that the analysis based on machine learning can be used to assist in personalized career advice by aligning the student strengths to the best job sectors. Such a strategy can assist students, educators and career counselors to make more informative decisions and contribute to the creation of AI-based counseling systems in the Bangladeshi universities.

A. Future Work

This study uncovers several important directions for more extensive research in the future. First, using a larger and more diverse dataset from different institutions, educational levels, and backgrounds would improve the applicability of the proposed model. In future work, we plan to incorporate k-fold cross-validation to obtain more robust and statistically reliable performance estimates. This will help reduce potential bias introduced by a single train–test split and improve the generalizability of the proposed multi-label career prediction model.

Second, we plan to explore advanced deep learning models such as recurrent networks, transformer-based systems, and attention-driven neural networks. These models, along with modern explainability methods, may help capture more complex patterns in student behavior and make the predictions easier to understand.

Third, future studies can look into feature selection and dimensionality reduction methods to reduce noise in the data and lessen the problems caused by imbalanced classes.

In addition, building a real-time or web-based system would allow students to easily receive personalized career suggestions. In the future, the system should be expanded into a longitudinal study that follows students over time. This would enable comparison of the predictions of the model and the actual job performance, and therefore the system would be more effective in career guidance.

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